1. Quality Goals & Standards

- **Definition of Quality:** For Streamline Retail, quality means delivering a secure, reliable, and highly usable mobile application that meets both internal stakeholder requirements and external user expectations. The software must align with business needs and perform flawlessly under real-world retail conditions.
- **Alignment with Business Goals:** Our primary focus is on customer retention through exceptional usability. Our solutions will gain user trust by being known for system stability and outstanding security.
- **Quality Metrics & Acceptance Thresholds:**
    - Achieve at least 80% test coverage for unit and integration testing.
    - Maintain a defect leakage rate of 2 or fewer reported production defects per month for issues with a severity higher than "Low".
    - Achieve a system uptime of 99.5%.
    - Ensure zero critical security incidents occur in the production environment.
    - Maintain a Customer Satisfaction (CSAT) rating of 90% or higher.

2. Process Integration

QA in Agile Ceremonies

- **Sprint Planning:** QA is responsible for estimating the testing effort, planning test activities, and confirming test data requirements. Additionally, QA reviews and refines the acceptance criteria alongside the Product Owner and the business analyst.
- **Daily Standups:** QA shares testing progress and blockers, and syncs with the engineers. In turn, engineers provide proactive updates on upcoming functionality that will be ready for testing that day.
- **Sprint Review:** QA demonstrates the testing coverage and reports on the test results as well as any outstanding defects.
- **Retrospective:** QA's role is to suggest actionable improvements to automation, processes, and team collaboration.

Embedded QA Activities in the SDLC

- **Unit Testing:** This activity is developer-led and requires at least 80% test coverage.
- **Code Reviews:** These are developer-led peer reviews that must also include a review of the tests.
- **Test Automation in CI/CD:** End-to-end test automation is QA-led and performed via low-code automation tools. The CI/CD pipeline automatically executes smoke, regression, and performance tests prior to every deployment.
- **Manual & Exploratory Testing:** Exploratory testing is conducted in every sprint to catch edge cases and User Experience (UX) issues.

RACI Matrix for Quality Activities

This matrix outlines the roles and responsibilities for key quality activities *(R = Responsible, A = Accountable, C = Consulted, I = Informed)*:

Activity	QA Engineer	Developer	Product Owner	Scrum Master

Write Unit Tests	C	RA	C	
Define Acceptance Criteria	R	A	C	
Execute Regression Tests	RA		C	
Triage Defects	R	RA	R	
Approve Release	C	A	C	R

3. Defect Management Process

Workflow & Responsibilities

- All defects are logged into the task tracking tool (e.g., Jira).
- A defect triaging huddle occurs daily to evaluate newly reported issues.
- In terms of responsibilities, QA assigns the **Severity** of the defect, while the Product Owner assigns the **Priority**.

Severity Definitions

- **Critical (S1):** Acceptance criteria from the task definition are not met. This includes instances where the visual appearance completely fails to match the designs, there is an immediate effect on the user's ability to execute primary scenarios, or there is an inability to perform major back-office functions with no workaround available. It also covers any production issue with an immediate effect on customer satisfaction or business continuity.
- **High (S2):** Acceptance criteria are only partially met. The layout might match the designs, but there are considerable flaws regarding accessibility or messaging. This severity also includes issues affecting user experience that do not quite classify as S1, or an S1 defect that has a temporary workaround.
- **Medium (S3):** Acceptance criteria are almost met. These are minor visual defects, insignificant functional flaws, typos in messaging, or issues that do not materially affect the performance or functionality of the solution.

- **Low (S4):** The issue raised is not covered by the acceptance criteria of the existing tasks. Therefore, it is considered an enhancement to the in-scope functionality and will be treated as a change request.

Service Level Agreements (SLAs) for Resolution

- **S1 (Critical):** Fix within 1 business day (with a Mean Time to Detect aiming for < 1 hour).
- **S2 (High):** Fix within 3 days (with a Mean Time to Detect aiming for < 1 day).
- **S3 (Medium):** Fix within 7 days unless it is deprioritized by the Product Owner.
- **S4 (Low):** Move to the project Backlog for future prioritization.

4. Verification & Validation

Systemic Quality Practices

- **Test-Driven Development (TDD):** A development cycle where requirements are turned into very specific test cases, and the software is improved specifically to pass those tests.
- **Behaviour-Driven Development (BDD):** An extension of TDD that converts structured natural language requirements into executable tests, closely linking acceptance criteria to validation tests.
- **3-Amigos Meetings:** Collaborative sessions used to align on the scope of acceptance, the amount of testing needed, and the design required to meet outcomes.

Testing Levels and Activities

- **Local Development:** Involves 3-amigos meetings, unit test creation, and ad-hoc dev-QA catch-ups for rapid validation. Development is only considered finished when unit tests pass, library code test coverage hits 100% where required, and the code passes review and merges successfully with automated regression passed.
- **System Integration Testing:** Test scenarios are updated when requirements are finalized. Testing is performed by QA, and the agreed-upon scenarios are automated.
- **User Acceptance Testing (UAT):** End-to-end testing is performed by QA once code is deployed to the UAT environment. This is followed by the execution of a regression test suite and an exploratory testing session. Finally, UAT is performed with selected users or the Product Owner.
- **Non-Functional Testing:** A specialized partner will be employed to perform penetration testing, security scans, and performance testing once per increment for a release candidate. Usability testing will commence as soon as the first visual prototypes are created.

Automation Strategy & Criteria

- 100% of unit tests and integration tests will be automated.
- End-to-end test cases will be automated using preferred low-code tools if they meet **all** of the following criteria: they show high repetition, are otherwise time-consuming to test manually, have algorithmically deterministic outcomes, and are based on stable requirements.
- Test cases will also be automated if they meet **at least one** of these secondary criteria: they rely on large data sets, involve cross-platform testing, or have UI elements.

5. Continuous Improvement

Feedback Loops and Process Refinement

- **Sprint Retrospectives:** These ceremonies will include a review of QA insights as a mandatory step, focusing specifically on elements like test failures and missed test scenarios.
- **Post-Release Reviews:** After a release, the team will perform reviews to thoroughly analyze any defect leakage and user complaints that occurred in production.
- **Root Cause Analysis:** The team will employ regular bug root cause analysis to identify specific gaps in coding or testing practices.

KPI Monitoring and Team Development

- **Monthly Review of QA KPIs:** Key Performance Indicators related to quality, such as test coverage, test cycle time, and defect leakage, will be reviewed on a monthly basis.
- **Training and Certification:** To ensure consistent application of quality standards, all project members will be provided with training and certification in Test-Driven Development (TDD), Behaviour-Driven Development (BDD), and the selected test automation tools.